

SHAWANO, WI, USA

WMO: 720589

Lat: 44.783N

Lon: 88.550W

Elev: 248

StdP: 98.38

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00185

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-19.0	-27.1	0.3	-20.7	-24.5	0.4	-18.4	8.6	-5.9	7.8	-5.3	1.4	270	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	31.4	22.3	30.0	21.6	28.6	20.8	24.1	28.8	23.2	27.8	22.2	26.6	3.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.9	27.0	21.8	17.0	26.0	20.8	16.0	25.3	73.9	29.4	70.0	27.8	66.4	26.8	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.2	6.1	DB	-25.1	34.4	2.1	1.4	-26.6	35.4	-27.9	36.2	-29.1	36.9	-30.6	37.9	
			WB	-25.4	25.8	2.1	1.5	-26.9	26.9	-28.1	27.7	-29.3	28.5	-30.9	29.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-7.0	-5.7	0.7	6.9	14.8	19.6	22.1	20.9	16.6	9.8	2.7	-3.5	(6)
	DBStd	11.33	6.51	6.05	6.30	4.85	4.86	3.51	2.99	2.94	4.21	4.43	5.23	5.73	(7)
	HDD10.0	2094	528	439	294	116	12	0	0	0	3	60	224	419	(8)
	HDD18.3	4012	786	672	546	343	132	26	4	9	81	266	469	677	(9)
	CDD10.0	1452	0	0	7	23	160	288	376	338	201	54	5	0	(10)
	CDD18.3	327	0	0	0	1	22	63	122	89	29	2	0	0	(11)
	CDH23.3	2846	0	0	6	16	258	541	1106	662	236	21	0	0	(12)
	CDH26.7	801	0	0	0	3	78	157	354	162	46	1	0	0	(13)
(14) Wind	WSAvg	2.4	2.4	2.6	2.6	3.2	2.9	2.4	2.0	1.8	2.1	2.5	2.4	2.4	(14)
(15) Precipitation	PrecAvg	800	31	24	44	71	96	106	99	89	87	69	47	37	(15)
	PrecMax	940	88	50	83	124	144	202	224	228	193	146	125	77	(16)
	PrecMin	675	7	6	5	14	42	45	28	17	21	5	6	4	(17)
	PrecStd	83	19	11	21	28	31	48	49	42	45	38	30	18	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	11.6	24.8	25.6	32.1	33.0	33.8	32.1	30.7	26.1	19.8	12.2	(19)
		MCWB	4.3	7.6	16.7	15.7	20.4	22.9	24.4	23.4	22.6	17.4	13.6	10.1	(20)
	2%	DB	4.8	7.4	17.5	21.4	28.8	30.4	31.5	30.1	27.9	22.7	15.3	7.3	(21)
		MCWB	2.7	4.9	11.8	13.4	18.4	20.8	23.1	22.4	20.8	15.7	10.2	5.4	(22)
	5%	DB	3.0	4.6	13.0	18.3	26.3	28.4	30.2	28.5	26.2	20.0	12.5	5.2	(23)
		MCWB	1.6	1.9	7.6	10.9	17.4	20.2	22.0	21.5	19.3	15.1	9.1	3.6	(24)
	10%	DB	1.8	2.5	9.8	15.8	23.7	26.6	28.7	27.2	24.2	17.3	10.1	3.5	(25)
		MCWB	0.7	0.5	5.9	9.1	16.2	18.9	21.1	20.3	19.1	13.2	7.5	2.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	8.0	17.2	18.1	22.2	24.8	26.4	24.8	23.6	20.2	14.2	10.5	(27)
		MCDB	6.7	10.7	24.8	23.1	28.7	30.8	32.6	29.2	28.0	22.9	18.2	11.6	(28)
	2%	WB	3.1	4.5	12.8	14.7	20.6	22.9	24.6	23.7	22.3	17.5	11.4	5.8	(29)
		MCDB	4.7	7.2	16.0	19.8	25.8	28.2	29.3	28.2	26.5	21.1	14.2	6.7	(30)
	5%	WB	1.8	2.3	8.5	11.8	19.1	21.7	23.4	22.7	21.1	15.6	9.3	3.9	(31)
		MCDB	2.8	4.2	13.1	17.4	24.1	25.9	27.9	26.9	24.7	19.4	12.1	5.3	(32)
	10%	WB	0.7	0.7	5.8	9.6	17.3	20.2	22.3	21.7	19.9	13.7	7.7	2.3	(33)
		MCDB	1.7	2.3	9.2	14.8	22.1	24.7	26.9	25.6	23.0	16.7	9.7	3.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	9.4	10.4	10.6	11.9	10.8	11.3	10.6	11.2	9.5	8.0	6.6	(35)
		MCDBR	8.0	10.2	14.3	15.9	15.4	12.9	12.6	11.6	13.3	13.9	12.4	8.0	(36)
	5% WB	MCWBR	6.7	7.5	9.1	9.2	7.1	6.1	6.0	5.1	6.7	8.2	8.1	6.6	(37)
		MCWBR	7.7	9.4	13.5	14.6	12.6	11.2	11.0	10.0	10.6	11.7	11.0	7.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.255	0.312	0.339	0.363	0.402	0.400	0.388	0.392	0.366	0.321	0.296	0.265	(40)
		taud	2.401	2.220	2.316	2.335	2.269	2.301	2.349	2.332	2.401	2.525	2.511	2.446	(41)
	Ebn at Noon		879	863	892	898	868	869	875	859	854	857	823	829	(42)
		Edh at Noon	72	105	111	119	131	128	120	118	101	78	66	63	(43)
(44) All-Sky Solar Radiation	RadAvg		1.61	2.70	3.87	4.56	5.47	5.93	6.10	5.19	4.10	2.58	1.62	1.17	(44)
	RadStd		0.17	0.26	0.40	0.46	0.44	0.44	0.30	0.34	0.30	0.26	0.15	0.12	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page